I. Chapter I: Foundations of Trustworthy AI

This Chapter sets out the foundations of Trustworthy AI, grounded in fundamental rights and reflected by four ethical principles that should be adhered to in order to ensure ethical and robust AI. It draws heavily on the field of ethics.

AI ethics is a sub-field of applied ethics, focusing on the ethical issues raised by the development, deployment and use of AI. Its central concern is to identify how AI can advance or raise concerns to the good life of individuals, whether in terms of quality of life, or human autonomy and freedom necessary for a democratic society.

Ethical reflection on AI technology can serve multiple purposes. First, it can stimulate reflection on the need to protect individuals and groups at the most basic level. Second, it can stimulate new kinds of innovations that seek to foster ethical values, such as those helping to achieve the UN Sustainable Development Goals[13], which are firmly embedded in the forthcoming EU Agenda 2030.[14] While this document mostly concerns itself with the first purpose mentioned, the importance that ethics could have in the second should not be underestimated. Trustworthy AI can improve individual flourishing and collective wellbeing by generating prosperity, value creation and wealth maximization. It can contribute to achieving a fair society, by helping to increase citizens' health and well-being in ways that foster equality in the distribution of economic, social and political opportunity.

It is therefore imperative that we understand how to best support AI development, deployment and use to ensure that everyone can thrive in an AI-based world, and to build a better future while at the same time being globally competitive. As with any powerful technology, the use of AI systems in our society raises several ethical challenges, for instance relating to their impact on people and society, decision-making capabilities and safety. If we are increasingly going to use the assistance of or delegate decisions to AI systems, we need to make sure these systems are fair in their impact on people's lives, that they are in line with values that should not be compromised and able to act accordingly, and that suitable accountability processes can ensure this.

Europe needs to define what normative vision of an AI-immersed future it wants to realise, and understand which notion of AI should be studied, developed, deployed and used in Europe to achieve this vision. With this document, we intend to contribute to this effort by introducing the notion of Trustworthy AI, which we believe is the right way to build a future with AI. A future where democracy, the rule of law and fundamental rights underpin AI systems and where such systems continuously improve and defend democratic culture will also enable an environment where innovation and responsible competitiveness can thrive.

A domain-specific ethics code – however consistent, developed and fine-grained future versions of it may be – can never function as a substitute for ethical reasoning itself, which must always remain sensitive to contextual details that cannot be captured in general Guidelines. Beyond developing a set of rules, ensuring Trustworthy AI requires us to build and maintain an ethical culture and mind-set through public debate, education and practical learning.

1. Fundamental rights as moral and legal entitlements

We believe in an approach to AI ethics based on the fundamental rights enshrined in the EU Treaties,[15] the EU Charter and international human rights law.[16] Respect for fundamental rights, within a framework of democracy and the rule of law, provides the most promising foundations for identifying abstract ethical principles and values, which can be operationalised in the context of AI.

The EU Treaties and the EU Charter prescribe a series of fundamental rights that EU member states and EU institutions are legally obliged to respect when implementing EU law. These rights are described in the EU Charter

13 https://ec.europa.eu/commission/publications/reflection-paper-towards-sustainable-europe-2030_en

14 https://sustainabledevelopment.un.org/?menu=1300

15 The EU is based on a constitutional commitment to protect the fundamental and indivisible rights of human beings, to ensure respect for the rule of law, to foster democratic freedom and promote the common good. These rights are reflected in Articles 2 and 3 of the Treaty on European Union, and in the Charter of Fundamental Rights of the EU.

16 Other legal instruments reflect and provide further specification of these commitments, such as for instance the Council of Europe's European Social Charter or specific legislation such as the EU's General Data Protection Regulation.

by reference to dignity, freedoms, equality and solidarity, citizens' rights and justice. The common foundation that unites these rights can be understood as rooted in respect for human dignity – thereby reflecting what we describe as a "human-centric approach" in which the human being enjoys a unique and inalienable moral status of primacy in the civil, political, economic and social fields.[17]

While the rights set out in the EU Charter are legally binding,[18] it is important to recognise that fundamental rights do not provide comprehensive legal protection in every case. For the EU Charter, for instance, it is important to underline that its field of application is limited to areas of EU law. International human rights law and in particular the European Convention on Human Rights are legally binding on EU Member States, including in areas that fall outside the scope of EU law. At the same time, fundamental rights are also bestowed on individuals and (to a certain degree) groups by virtue of their moral status as human beings, independently of their legal force. Understood as legally enforceable rights, fundamental rights therefore fall under the first component of Trustworthy AI (lawful AI), which safeguards compliance with the law. Understood as the rights of everyone, rooted in the inherent moral status of human beings, they also underpin the second component of Trustworthy AI (ethical AI), dealing with ethical norms that are not necessarily legally binding yet crucial to ensure trustworthiness. Since this document does not aim to offer guidance on the former component, for the purpose of these non-binding guidelines, references to fundamental rights reflect the latter component.

2. From fundamental rights to ethical principles

2.1 Fundamental rights as a basis for Trustworthy AI

Among the comprehensive set of indivisible rights set out in international human rights law, the EU Treaties and the EU Charter, the below families of fundamental rights are particularly apt to cover AI systems. Many of these rights are, in specified circumstances, legally enforceable in the EU so that compliance with their terms is legally obligatory. But even after compliance with legally enforceable fundamental rights has been achieved, ethical reflection can help us understand how the development, deployment and use of AI systems may implicate fundamental rights and their underlying values, and can help provide more fine-grained guidance when seeking to identify what we *should* do rather than what we (currently) *can* do with technology.

Respect for human dignity. Human dignity encompasses the idea that every human being possesses an "intrinsic worth", which should never be diminished, compromised or repressed by others – nor by new technologies like AI systems.[19] In this context, respect for human dignity entails that all people are treated with respect due to them as moral *subjects*, rather than merely as *objects* to be sifted, sorted, scored, herded, conditioned or manipulated. AI systems should hence be developed in a manner that respects, serves and protects humans' physical and mental integrity, personal and cultural sense of identity, and satisfaction of their essential needs.[20]

Freedom of the individual. Human beings should remain free to make life decisions for themselves. This entails freedom from sovereign intrusion, but also requires intervention from government and non-governmental organisations to ensure that individuals or people at risk of exclusion have equal access to AI's benefits and opportunities. In an AI context, freedom of the individual for instance requires mitigation of (in)direct illegitimate coercion, threats to mental autonomy and mental health, unjustified surveillance, deception and unfair manipulation. In fact, freedom of the individual means a commitment to enabling individuals to wield even higher control over their lives, including (among other rights) protection of the freedom to conduct a business, the freedom of the arts and science, freedom of expression, the right to private life and privacy, and freedom of

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17 It should be noted that a commitment to human-centric AI and its anchoring in fundamental rights requires collective societal and constitutional foundations in which individual freedom and respect for human dignity is both practically possible and meaningful, rather than implying an unduly individualistic account of the human.

18 Pursuant to Article 51 of the Charter, it applies to EU Institutions and to EU member states when implementing EU law.

19 C. McCrudden, Human Dignity and Judicial Interpretation of Human Rights, *EJIL*, 19(4), 2008.

20 For an understanding of "human dignity" along these lines see E. Hilgendorf, Problem Areas in the Dignity Debate and the Ensemble Theory of Human Dignity, in: D. Grimm, A. Kemmerer, C. Möllers (eds.), *Human Dignity in Context. Explorations of a Contested Concept*, 2018, pp. 325 ff.

assembly and association**.**

Respect for democracy, justice and the rule of law. All governmental power in constitutional democracies must be legally authorised and limited by law. AI systems should serve to maintain and foster democratic processes and respect the plurality of values and life choices of individuals. AI systems must not undermine democratic processes, human deliberation or democratic voting systems. AI systems must also embed a commitment to ensure that they do not operate in ways that undermine the foundational commitments upon which the rule of law is founded, mandatory laws and regulation, and to ensure due process and equality before the law.

Equality, non-discrimination and solidarity *- including the rights of persons at risk of exclusion.* Equal respect for the moral worth and dignity of all human beings must be ensured. This goes beyond non-discrimination, which tolerates the drawing of distinctions between dissimilar situations based on objective justifications. In an AI context, equality entails that the system's operations cannot generate unfairly biased outputs (e.g. the data used to train AI systems should be as inclusive as possible, representing different population groups). This also requires adequate respect for potentially vulnerable persons and groups,[21] such as workers, women, persons with disabilities, ethnic minorities, children, consumers or others at risk of exclusion.

Citizens' rights. Citizens benefit from a wide array of rights, including the right to vote, the right to good administration or access to public documents, and the right to petition the administration. AI systems offer substantial potential to improve the scale and efficiency of government in the provision of public goods and services to society. At the same time, citizens' rights could also be negatively impacted by AI systems and should be safeguarded. When the term "citizens' rights" is used here, this is not to deny or neglect the rights of third-country nationals and irregular (or illegal) persons in the EU who also have rights under international law, and – therefore – in the area of AI systems.

2.2 Ethical Principles in the Context of AI Systems[22]

Many public, private, and civil organizations have drawn inspiration from fundamental rights to produce ethical frameworks for AI systems.[23] In the EU, the European Group on Ethics in Science and New Technologies ("EGE") proposed a set of 9 basic principles, based on the fundamental values laid down in the EU Treaties and Charter.[24] We build further on this work, recognising most of the principles hitherto propounded by various groups, while clarifying the ends that all principles seek to nurture and support. These ethical principles can inspire new and specific regulatory instruments, can help interpreting fundamental rights as our socio-technical environment evolves over time, and can guide the rationale for AI systems' development, deployment and use – adapting dynamically as society itself evolves.

AI systems should improve individual and collective wellbeing. This section lists **four ethical principles**, rooted in fundamental rights, which must be respected in order to ensure that AI systems are developed, deployed and used in a trustworthy manner. They are specified as **ethical imperatives**, such that AI practitioners should always strive to adhere to them. Without imposing a hierarchy, we list the principles here below in manner that mirrors the order of appearance of the fundamental rights upon which they are based in the EU Charter.[25]

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21 For a description of the term as used throughout this document, see the Glossary.

22 These principles also apply to the development, deployment and use of other technologies, and hence are not specific to AI systems. In what follows, we have aimed to set out their relevance specifically in an AI-related context.

23 Reliance on fundamental rights also helps to limit regulatory uncertainty as it can build on the basis of decades of practice of fundamental rights protection in the EU, thereby offering clarity, readability and foreseeability.

24 More recently, the AI4People's taskforce has surveyed the aforementioned EGE principles as well as 36 other ethical principles put forward to date and subsumed them under four overarching principles: L. Floridi, J. Cowls, M. Beltrametti, R. Chatila, P. Chazerand, V. Dignum, C. Luetge, R. Madelin, U. Pagallo, F. Rossi, B. Schafer, P. Valcke, E. J. M. Vayena (2018), "AI4People —An Ethical Framework for a Good AI Society: Opportunities, Risks, Principles, and Recommendations", Minds and Machines 28(4): 689-707.

25 Respect for human autonomy is strongly associated with the right to human dignity and liberty (reflected in Articles 1 and 6 of the Charter). The prevention of harm is strongly linked to the protection of physical or mental integrity (reflected in Article 3).

These are the principles of:

(i) Respect for human autonomy
(ii) Prevention of harm
(iii) Fairness
(iv) Explicability

Many of these are to a large extent already reflected in existing legal requirements for which mandatory compliance is required and hence also fall within the scope of lawful AI, which is Trustworthy AI's first component.[26] Yet, as set out above, while many legal obligations reflect ethical principles, adherence to ethical principles goes beyond formal compliance with existing laws.[27]

- *The principle of respect for human autonomy*

The fundamental rights upon which the EU is founded are directed towards ensuring respect for the freedom and autonomy of human beings. Humans interacting with AI systems must be able to keep full and effective self-determination over themselves, and be able to partake in the democratic process. AI systems should not unjustifiably subordinate, coerce, deceive, manipulate, condition or herd humans. Instead, they should be designed to augment, complement and empower human cognitive, social and cultural skills. The allocation of functions between humans and AI systems should follow human-centric design principles and leave meaningful opportunity for human choice. This means securing human oversight[28] over work processes in AI systems. AI systems may also fundamentally change the work sphere. It should support humans in the working environment, and aim for the creation of meaningful work.

- *The principle of prevention of harm*

AI systems should neither cause nor exacerbate harm[29] or otherwise adversely affect human beings.[30] This entails the protection of human dignity as well as mental and physical integrity. AI systems and the environments in which they operate must be safe and secure. They must be technically robust and it should be ensured that they are not open to malicious use. Vulnerable persons should receive greater attention and be included in the development, deployment and use of AI systems. Particular attention must also be paid to situations where AI systems can cause or exacerbate adverse impacts due to asymmetries of power or information, such as between employers and employees, businesses and consumers or governments and citizens. Preventing harm also entails consideration of the natural environment and all living beings.

- *The principle of fairness*

The development, deployment and use of AI systems must be fair. While we acknowledge that there are many different interpretations of fairness, we believe that fairness has both a substantive and a procedural dimension. The substantive dimension implies a commitment to: ensuring equal and just distribution of both benefits and costs, and ensuring that individuals and groups are free from unfair bias, discrimination and stigmatisation. If unfair biases can be avoided, AI systems could even increase societal fairness. Equal opportunity in terms of access to education, goods, services and technology should also be fostered. Moreover, the use of AI systems should never lead to people being deceived or unjustifiably impaired in their freedom of choice. Additionally, fairness implies that AI practitioners should respect the principle of proportionality between means and ends, and consider carefully how to

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Fairness is closely linked to the rights to Non-discrimination, Solidarity and Justice (reflected in Articles 21 and following). Explicability and Responsibility are closely linked to the rights relating to Justice (as reflected in Article 47).

26 Think for instance of the GDPR or EU consumer protection regulations.

27 For further reading on this subject, see for instance L. Floridi, Soft Ethics and the Governance of the Digital, *Philosophy & Technology,* March 2018, Volume 31, Issue 1, pp 1–8.

28 The concept of human oversight is further developed as one of the key requirements set out in Chapter II here below.

29 Harms can be individual or collective, and can include intangible harm to social, cultural and political environments.

30 This also encompasses the way of living of individuals and social groups, avoiding for instance cultural harm.

balance competing interests and objectives.[31] The procedural dimension of fairness entails the ability to contest and seek effective redress against decisions made by AI systems and by the humans operating them.[32] In order to do so, the entity accountable for the decision must be identifiable, and the decision-making processes should be explicable.

- *The principle of explicability*

Explicability is crucial for building and maintaining users' trust in AI systems. This means that processes need to be transparent, the capabilities and purpose of AI systems openly communicated, and decisions – to the extent possible – explainable to those directly and indirectly affected. Without such information, a decision cannot be duly contested. An explanation as to why a model has generated a particular output or decision (and what combination of input factors contributed to that) is not always possible. These cases are referred to as 'black box' algorithms and require special attention. In those circumstances, other explicability measures (e.g. traceability, auditability and transparent communication on system capabilities) may be required, provided that the system as a whole respects fundamental rights. The degree to which explicability is needed is highly dependent on the context and the severity of the consequences if that output is erroneous or otherwise inaccurate.[33]

2.3 Tensions between the principles

Tensions may arise between the above principles, for which there is no fixed solution. In line with the EU fundamental commitment to democratic engagement, due process and open political participation, methods of accountable deliberation to deal with such tensions should be established. For instance, in various application domains, *the principle of prevention of harm* and *the principle of human autonomy* may be in conflict. Consider as an example the use of AI systems for 'predictive policing', which may help to reduce crime, but in ways that entail surveillance activities that impinge on individual liberty and privacy. Furthermore, AI systems' overall benefits should substantially exceed the foreseeable individual risks. While the above principles certainly offer guidance towards solutions, they remain abstract ethical prescriptions. AI practitioners can hence not be expected to find the right solution based on the principles above, yet they should approach ethical dilemmas and trade-offs via reasoned, evidence-based reflection rather than intuition or random discretion.

There may be situations, however, where no ethically acceptable trade-offs can be identified. Certain fundamental rights and correlated principles are absolute and cannot be subject to a balancing exercise (e.g. human dignity).

Key guidance derived from Chapter I:

- ✓ Develop, deploy and use AI systems in a way that adheres to the ethical principles of: *respect for human autonomy, prevention of harm, fairness and explicability.* Acknowledge and address the potential tensions between these principles.
- ✓ Pay particular attention to situations involving more vulnerable groups such as children, persons with disabilities and others that have historically been disadvantaged or are at risk of exclusion, and to situations which are characterised by asymmetries of power or information, such as between employers and workers, or between businesses and consumers.[34]

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31 This is relates to the principle of proportionality (reflected in the maxim that one should not 'use a sledge hammer to crack a nut'). Measures taken to achieve an end (e.g. the data extraction measures implemented to realise the AI optimisation function) should be limited to what is strictly necessary. It also entails that when several measures compete for the satisfaction of an end, preference should be given to the one that is least adverse to fundamental rights and ethical norms (e.g. AI developers should always prefer public sector data to personal data). Reference can also be made to the proportionality between user and deployer, considering the rights of companies (including intellectual property and confidentiality) on the one hand, and the rights of the user on the other.

32 Including by using their right of association and to join a trade union in a working environment, as provided for by Article 12 of the EU Charter of fundamental rights.

33 For example, little ethical concern may flow from inaccurate shopping recommendations generated by an AI system, in contrast to AI systems that evaluate whether an individual convicted of a criminal offence should be released on parole.

34 See articles 24 to 27 of the EU Charter, dealing with the rights of the child and the elderly, the integration of persons with disabilities and workers' rights. See also article 38 dealing with consumer protection.